Bail(ey)ing Out: Modeling Considerations and Sources

Working notes for CMCM 2026 (Bailey Hall egress). Each section covers a factor that affects egress time, what the research says about it, and how it can enter our model. Every figure links to the paper or page it came from. Re-check each number against the original source before it goes into the final manuscript.

Starting assumption: delay, not panic

Researchers who study real emergencies agree that "panic" is not an accurate description of crowd behavior. Crowds tend to behave better than expected, and cooperation is common (Cocking, isurv). Most lost time comes from people starting to move late. The model should represent delay, habit, imitation and cooperation, not irrational panic.

Behavior (decisions)

Response time before moving

This is probably the parameter with the largest effect on total egress time.

- An analysis of 40 unannounced evacuations (2,486 data points, six occupancy types) found that pre-movement times are usually well described by a lognormal or log-logistic distribution (Forssberg et al., 2019).
- In an unannounced evacuation of about 1,200 people during a live performance at the Marlowe Theatre, response time increased with the seat's distance from the exit aisles and the row's distance from an exit (Galea et al., 2017). Row position matters in the reaction phase, not only during movement.
- In university buildings where students use electronic devices, pre-movement times were very high and strongly linked to attachment to belongings (D'Orazio and Bernardini). This applies to the lecture case (Q3).
- Unannounced cinema evacuations showed social influence during the initial phase (Nilsson and Johansson, 2009). A review of VR studies reports that people who are physically closer have more influence, and that social influence matters more when the alarm cue is unclear (VR decision-making review, arXiv 2302.10339).

Model: a lognormal response time whose median increases with distance to the aisle. A Crowd Manager announcement shifts the distribution down. The probability of starting to move in each time step increases with the fraction of neighbors who are already moving.

Exit choice: shortest route, habit, imitation and queue avoidance

- Habit: in a mock furniture-store experiment, about 71% of occupants left through the door they had used before, and people tended to use the main entrance unless signage gave them other information (VR exit choice study, arXiv 2110.11577).
- Habit and imitation interact: in a VR fire evacuation, participants preferred the familiar exit,

more so when virtual neighbors went there. When the neighbors went to the unfamiliar exit, the familiarity effect disappeared (Kinateder and Warren, 2017).
- Queue avoidance: in a large crowd, participants avoided the more crowded exit when the doors were narrow, but not when they were wide (Kinateder and Warren, 2021).

Model: agent types (shortest-time, habitual, follower), each with its own multinomial logit weights. The share of each type is a scenario parameter. The queue-avoidance term should depend on door width.

Politeness versus pushing

- Faster-is-slower: a crowd that is more eager to leave can take longer, because clogs form at the door (Nicolas, Bouzat and Kuperman, 2016).
- In the same group's experiments with moderate competitiveness, flow rate rose steadily with the fraction of selfish participants, an effect they call faster-is-faster (same source).
- Measured flows: about 1.9 persons per second per meter of door width under normal conditions (Seyfried et al., as cited by Nicolas et al., 2016), and 3.3 to 3.7 per meter in competitive experiments with 70 to 80 cm doors (Nicolas et al., arXiv 1610.05909).
- A spatial game model shows that threatening conditions raise the proportion of impatient agents, which causes clogging and lower flow at bottlenecks even when smooth flow would be possible (Heliövaara et al., 2013).

Model: door capacity is a function of urgency. It rises with moderate urgency and drops with extreme pushing. The SFPE Handbook design value of about 1.3 persons/s/m is conservative and works as a lower bound (not linked here; check the egress chapter in a library copy).

Social groups

- People in social groups took longer to start moving and longer to reach the area near the exits, but there was no measurable difference directly in front of the exits (Bode et al., 2015).
- A laboratory study with pupils found, unexpectedly, that evacuation time decreased as group size increased (von Krüchten et al., 2016).

Model: a group chooses one exit together, starts when all members are ready, and moves at the pace of its slowest member. The case to watch is a separated group (for example during intermission): members move toward each other before leaving, which creates counterflow.

Helping behavior

In simulated evacuations, people with a strong shared identity tended to help a fallen person even though it delayed their own exit (Drury et al., University of Sussex). A fall therefore delays the helpers as well as the person who fell.

Movement

Position in the row

Beyond response time, each row has to clear: the person in seat k waits for the k − 1 people ahead. Predtechenskii and Milinskii give a maximum speed of 0.45 m/s along seat rows, and a

network model of lecture theatres gave reasonable predictions once exit-door and aisle constrictions were included (Lecture theatre network model, Fire Safety Science 10). On the Bailey seating chart, labels such as "18-20" appear to mean 18 chairs of 20 inches; confirm this before using it. Model: each row is a small queue that feeds the aisle with an alternating (zipper) merge.

Age and walking speed

- People over 65 walk about 20 to 25% slower than adults aged 18 to 40, and slower people determine flow rates (Thompson, citing Ando et al., 1988).
- Unimpeded stair descent: mean of about 0.71 m/s for adults and about 0.50 m/s for elderly people (SFPE Europe, anthropometric data and movement speeds).

Model: draw walking speeds from a different distribution for each scenario (an older audience at a classical concert, students at a lecture).

Luggage, backpacks and coats

In a bottleneck experiment, going from 0% to 100% luggage-carrying pedestrians cut mean flow by 39% in normal conditions and 47% in emergency conditions, and steady velocity by 22% and 33% (Deng et al., Transportmetrica A). These were suitcases, so backpacks should have a smaller effect. Winter coats matter in Ithaca, and in an ordinary departure (Q1) people put them on before leaving. Model: a capacity reduction factor plus extra response time.

Wheelchair users

In a bottleneck experiment, wheelchair users, and to a lesser degree people with luggage, needed longer to pass, which also slowed the people behind them (Together apart, arXiv 2311.01236). Bailey has up to three wheelchair positions and the elevator is prohibited during evacuation. Model them as slow, wider agents, or state this as a limitation (area of refuge with assistance).

Falls

- In a cellular automaton stampede model, more pedestrians fell as density increased (Huo et al., Physica A, 2022).
- Video analysis of the Running of the Bulls found that a fallen person sharply raises the falling probability of people nearby, and that the product of the local density gradient and walking speed indicates an imminent fall (arXiv 2211.08456).

Model: a per-step fall probability that depends on local density and speed, higher on stairs. A fallen agent blocks a lane for a random time. Use this as a stress scenario.

Density as a safety metric

Report the peak density at each node, not only the egress time. A plan that saves 10 seconds but creates dangerous density at the foyer doors is a worse plan.

Counterflow

People going back for belongings or family members, and responders coming in, reduce corridor capacity. Corridor barriers reduce counterflow.

Environment and information

Signage and LED compliance

- Only 38% of people perceive conventional exit signs (Galea et al., 2016).
- Exit signs with flashing green LEDs reached detection rates of up to 77% in experiments (Filippidis et al., 2021).
- In full-scale trials, an improved dynamic signage system, paired with voice messages that did not contradict the signs, redirected 66% of participants to the intended exit (Galea et al., improved ADSS trials).

Model consequence: LED guidance does not mean full compliance. Use a compliance probability of about 0.6 to 0.7 as the base case and sweep it. Crowd Manager instructions must agree with the LEDs, and a contradiction between the two is a failure mode worth simulating.

Smoke and darkness

- Evacuation models usually take the relation between smoke density and walking speed from one of two datasets, Jin's or Frantzich and Nilsson's (Ronchi et al.).
- Frantzich and Nilsson tested flashing lights at emergency exits and directional guide lights in a smoke-filled tunnel, which is close to our LED intervention (Frantzich and Nilsson, 2003).

At a concert the house lights are dimmed when the alarm sounds, which adds recognition delay.

Blocked exits

Experiments in a narrow seated area studied unpredictable failures of the exits at the ends of a long aisle, including how people switch exits and the effect of explicit guidance (J. Stat. Mech. seated-area experiment). Model: agents discover a blocked exit when they reach it, or learn about it from a Crowd Manager broadcast after a delay.

Training and staff

- In the lecture-theatre study, students were assumed to know the procedure because trial evacuations ran every six months (Fire Safety Science 10).
- Three announced drills in a theatre with different levels of staff intervention provide route-choice data for calibrating the Crowd Manager effect (Fire Technology, 2018).

Scenarios

Run about a dozen named scenarios and use Monte Carlo over the continuous parameters, instead of every possible combination.

Varying occupancy: plot total egress time T(N) against the number of occupants. At low N, response and walking time dominate. At high N, bottleneck capacity dominates. The knee of the curve shows whether the 180-second requirement is limited by response or by capacity, and the

slope after the knee equals one over the capacity of the minimum cut. With fewer attendees the law also requires fewer Crowd Managers (600 people need 3).

- Ordinary departure after a concert (Q1).
- Concert emergency with house lights down, alarm and Crowd Managers (Q2 base case).
- One exit blocked, each exit in turn, known versus unknown.
- Fire near the stage: stage routes and north balcony stairs unusable, smoke in the auditorium.
- Fire or smoke in the foyer: the main route is lost. LEDs that can mark an exit as closed help most here.
- Intermission: people in the foyer and restrooms, separated groups, counterflow back inside.
- Lecture (Q3): backpacks, laptops, possible doubt about whether the alarm is real.
- Occupancy at 25, 50, 75 and 100%, plus uneven distributions (full balcony, empty orchestra).
- Winter: ice or snow on the monumental staircase and portico stairs, coats, and people stopping just outside the doors.
- Power failure.
- One Crowd Manager absent or out of position.
- A fall on the balcony stairs.
- Older versus younger audience.
- Out of scope, for the limitations section: active threat, where the protocol (shelter versus evacuate) is different.

Markov chains

Markov chains are useful as components of the model, not as the whole model.

Behavioral state machine

Each person moves through states: seated and unaware, aware, ready (belongings collected, group together), walking, queuing, and exited (absorbing). Transition probabilities per time step can depend on neighbors, for example P(aware to ready) increasing with the share of neighbors already moving. This reproduces social influence and produces response-time distributions similar to the measured lognormal shapes.

Absorbing chain for one person's route

Graph nodes are states and exits are absorbing states. Let Q be the transient-to-transient block and R the transient-to-absorbing block. Then N = (I − Q)^(−1) is the fundamental matrix, t = N·1 gives the expected number of steps to exit, and B = N·R gives the probability that each zone ends at each exit. This gives expected exit loads analytically for any choice model, without simulation, but it ignores congestion. Deng et al. (CDC 2008) used this structure: a single-agent Markov transition matrix that ignores congestion, combined with queues at each building node.

Why not make the whole model a Markov chain

If the state is the position of every person, the state space explodes: in Deng et al.'s example,

one agent gives 255 states, but one agent per node gives 2^255 (Deng et al., CDC 2008). Our simulation is still a Markov chain if each update depends only on the current state, and the Monte Carlo runs sample it. That is worth one sentence in the paper as formal framing.

Precedent

A discrete-time Markov chain model for high-rise evacuation, including merging at stairs and floor congestion, gave results consistent with the SFPE method, the NIST egress simulator and an agent-based model (Physica A, 2024). In that formulation the state is the occupancy of each node, which is close to our queue network.

Braess paradox

Adding a route can increase total time when each person chooses selfishly, and removing an option can reduce it. Corridor barriers (Q4a) follow this logic: without barriers, the corridor lets everyone converge on the foyer.
The evidence is mixed. A 2025 study that combined supervised experiments with traffic-assignment models concluded that Braess' paradox is unlikely to be common in pedestrian evacuation. With autonomous evacuation and full network knowledge it did not appear in their high-demand cases, which points to information conditions as the deciding factor (Wang et al., 2025).
For Bailey: in the lecture case (full information), pure Braess may not appear. In the concert case, the foyer overload comes from habit and missing information rather than from Braess in the strict sense. Barriers address both.
Test it on the network by computing three outcomes:

1. User equilibrium (Wardrop): each person picks the lowest time given everyone else's choices.
2. System optimum: the lowest possible time for the last person out.
3. Habit outcome: each person uses the habitual exit.

Report the price of anarchy, PoA = T_UE / T_SO. Then remove the corridor links (barriers) and recompute the user equilibrium. If T_UE falls, that is Braess in Bailey. If it does not, the gain from barriers comes from information and coordination. Either result is worth reporting.
Small-scale version: Hughes suggested that well-placed obstacles can increase the flow through an exit, an inversion of Braess' paradox (Macroscopic modeling and simulations of room evacuation, arXiv 1308.1770). A conservation-law model reproduced a lower evacuation time only for specific obstacle positions, and a badly placed obstacle congested quickly (arXiv 1503.02826). Run a sensitivity analysis on barrier position.

Differential games and mean field games

Each evacuee minimizes their own time, and their cost depends on everyone else, so egress is a game. A differential game with 1,329 players cannot be solved directly.

- The large-population limit is a mean field game. Lachapelle and Wolfram model pedestrians who interact rationally and anticipate the future, which gives a forward-backward system in time: a Hamilton-Jacobi-Bellman equation solved backward for each person's optimal cost, and a Fokker-Planck equation solved forward for crowd density (Lachapelle and Wolfram, 2011).

- Burger, Di Francesco, Markowich and Wolfram present a mean field framework that unifies several evacuation models and relates to Hughes' eikonal model (arXiv 1304.5201).
- A mean field game for multilevel building evacuation has agents avoid dense areas to reduce walking and queuing costs at stairs and exits (multilevel evacuation MFG).

Conceptual payoff: the mean field game equilibrium is a Nash equilibrium of selfish agents, close to the lecture case. Mean field control is the social planner, close to the plan that barriers and LEDs impose. The gap between the two is the price of anarchy, which is the theory behind Q4.

For a weekend, use discrete versions inside the network model:

- Exit choice as a congestion game: each agent switches to the exit with the lowest estimated time given the others, repeated until no one switches. This game has a pure-strategy Nash equilibrium, with an upper bound on the number of iterations needed (Ehtamo et al., 2010).
- Patient versus impatient behavior at doors (the hawk-dove game of Heliövaara et al., 2013) justifies the capacity-drop parameter.
- Mean field games: a paragraph in the discussion, or a 1D corridor model if a teammate wants to work with PDEs.

Caveat: a known limitation of game models for exit choice is the assumption that evacuees act rationally (Ibrahim et al.). The game approach fits the trained lecture audience better than the concert audience, where habit dominates.

What goes into the model

Factor	How it enters the model	Priority
Response time	Lognormal that depends on seat position, or Markov states	Essential
Exit choice	Agent types plus multinomial logit	Essential
Door and stair capacity	Flow per meter, with a range that depends on urgency	Essential
Position in the row	Row queue plus zipper merge into the aisle	Essential
Compliance with Crowd Managers and LEDs	Calibrated probability (about 0.6 to 0.7 for LEDs)	High (Q4)
Blocked exit	Per-agent knowledge, discovery versus broadcast	High (Q2)
Age and walking speed	Speed distribution per scenario	High
Social groups	Joint exit choice, waiting for the slowest member	Medium
Luggage and coats	Capacity factor plus longer	Medium

	response time	
Falls	Stress scenario	Medium
Counterflow	Intermission scenario	Medium
Smoke and darkness	Speed reduction	Low (discussion)
Wheelchair users	Limitation plus area of refuge	Discussion
Mean field games	Theoretical framing	Discussion

Each added factor needs calibration. A tornado sensitivity chart will show which parameters change the 95th-percentile egress time. Response time, door capacity and compliance are the likely leaders, and that ranking is itself a useful finding for the executive summary.

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